Benign Tumors and Fibrocystic Changes.

By

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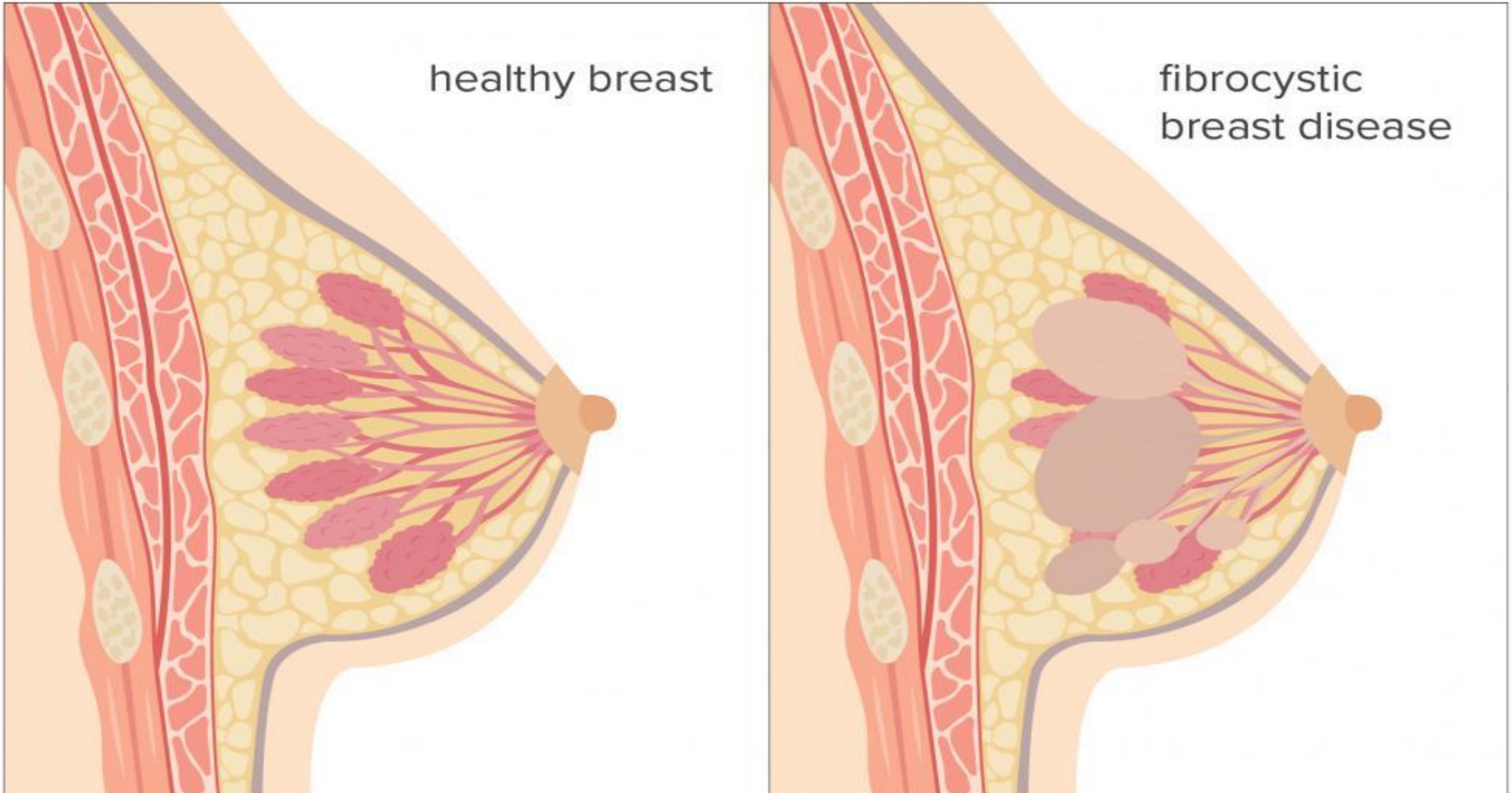
Fibrocystic changes

- Development of fibrosis and cysts
- Most common change in premenopausal breast
- Presents as lumpy breast, usually in UOQ
- Cysts have blue-domed appearance on gross

Benign, some changes carry increased risk for invasive carcinoma

- Fibrosis, cysts, and apocrine metaplasia
- Ductal hyperplasia and sclerosing adenosis (2x)
- Atypical hyperplasia (5x)

Fibrocystic Breast Disease



Intraductal Papiloma

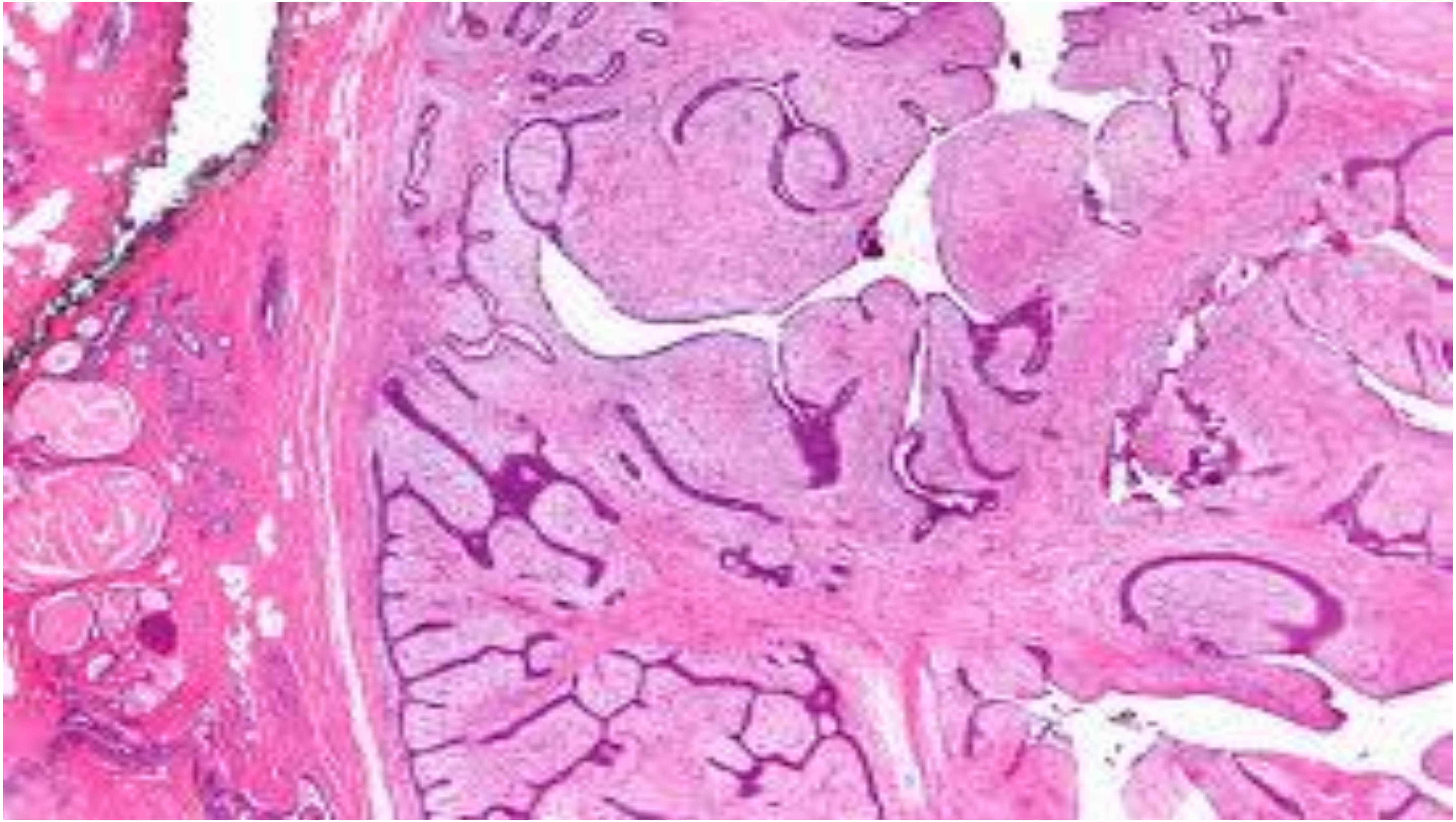
- Papillary growth, usually into large duct
- Fibrovascular projections and lined by epithelial and myoepithelial cells.
- Bloody nipple discharge in premenopausal woman
- Must be distinguished from papillary carcinoma

Fibroadenoma

- Tumor of fibrous tissue and glands
- Most common benign neoplasm of breast
- Well- circumscribed, mobile marble like mass
- Estrogen sensitive
- Benign, no increased risk for carcinoma

Phyllodes tumor

- Fibroadenoma –like tumor with overgrowth of fibrous component
- Leave like projections
- Most commonly seen in postmenopausal women
- Can be malignant



BREAST CANCER

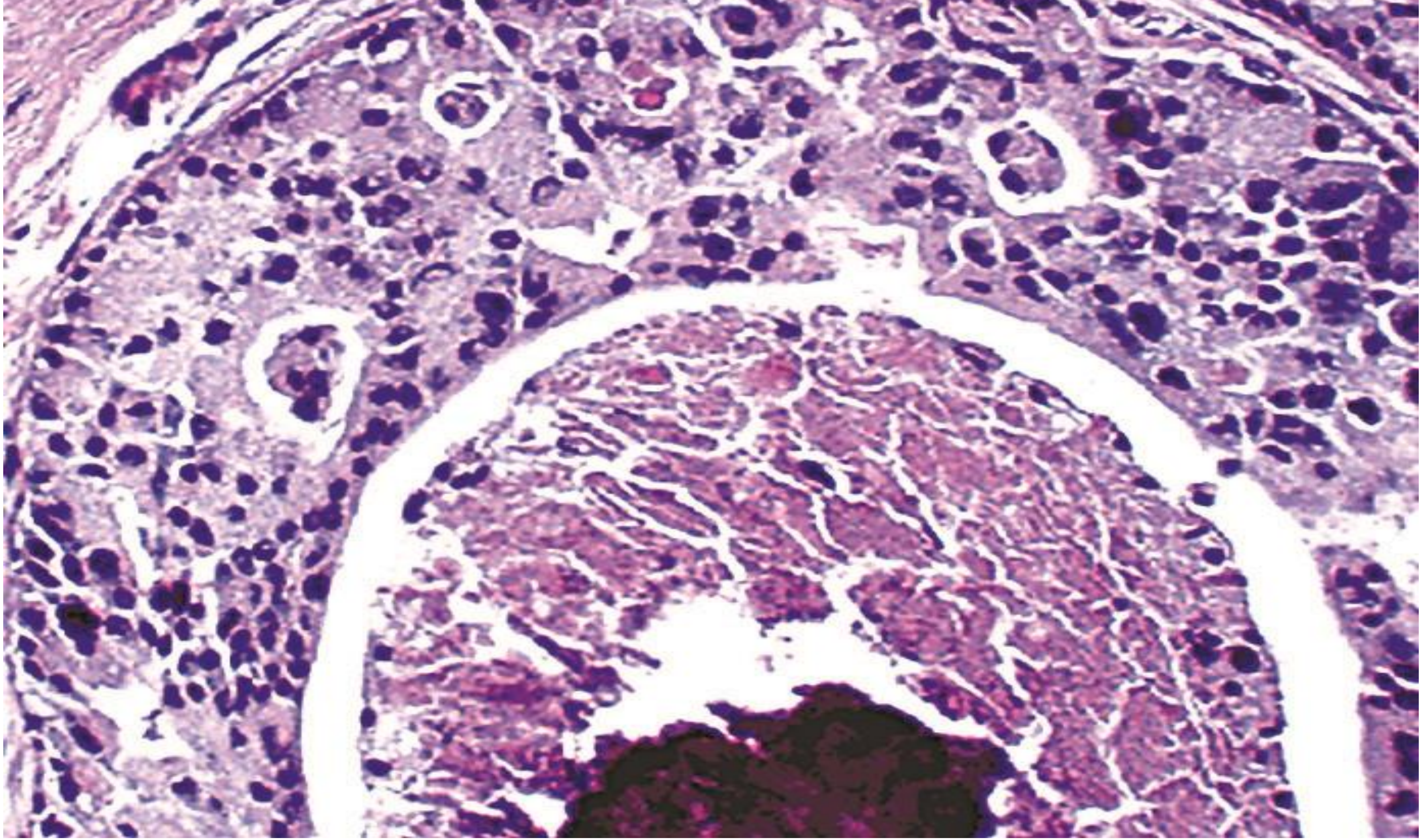
- Most common carcinoma in women by incidence
- Second most common cause of cancer mortality
- Risk factors mostly related to estrogen exposure

Risk Factors

- Female gender
- Age
- Early Menarch / late menopause
- Obesity
- Atypical hyperplasia
- First degree relative with breast cancer

DCIS

- Malignant proliferation of cells in ducts
- No invasion of basement membrane
- Detected as calcification on mammography
- Comedo type has high grade cells with necrosis and dystrophic calcification in the center of ducts



Paget Disease

- DCIS that extends ducts to skin of the nipple
- Presents as nipple ulceration and erythema
- Almost always associated with underlying carcinoma

Invasive Ductal Carcinoma

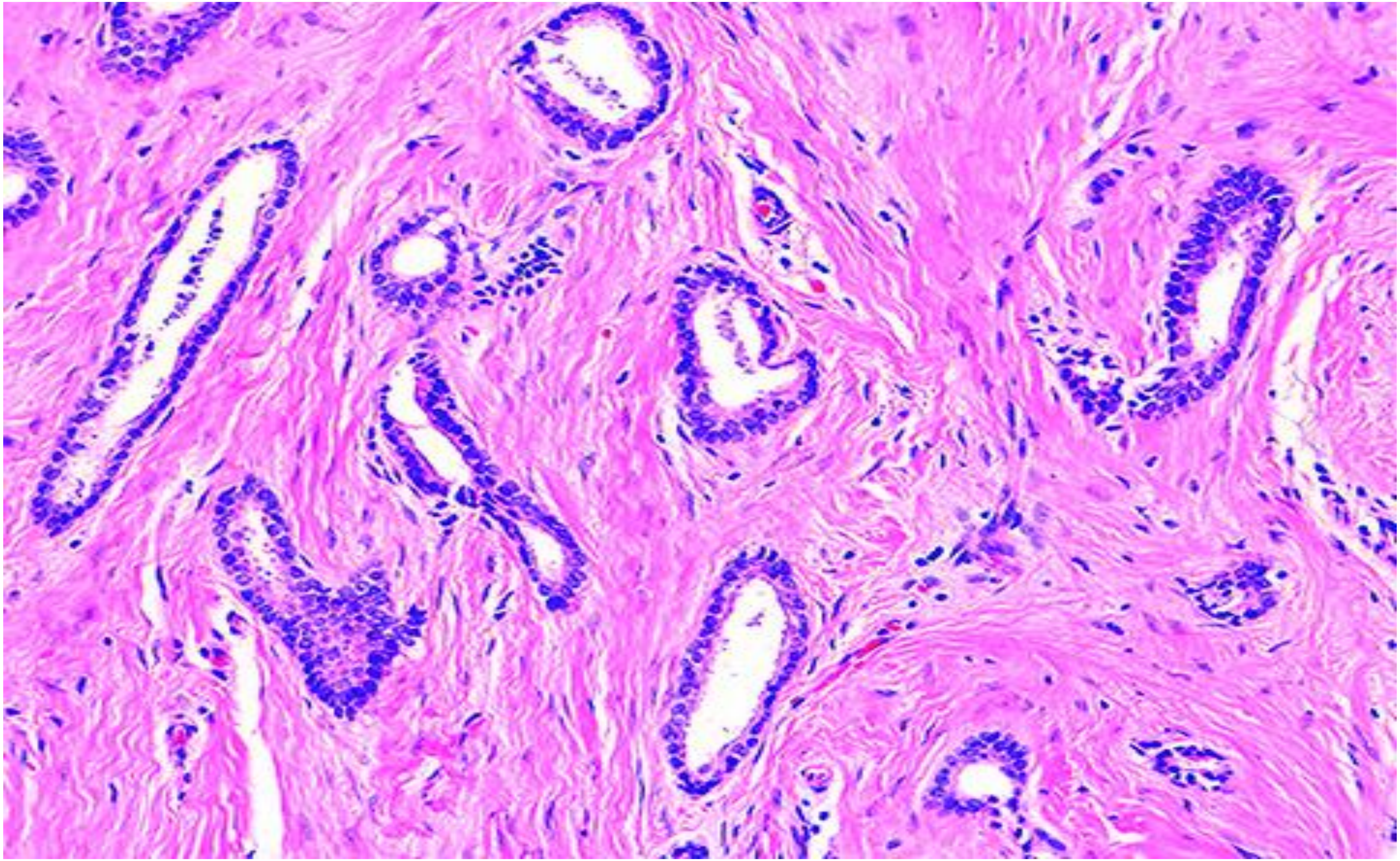
- Classically forms duct – like structures
- Most common type of invasive carcinoma
- Presents as a mass detected by physical exams or mammograghy
- Advanced tumors may result in dimpling of skin or retraction of nipple

Biopsy

- Duct like structures in desmoplastic stroma

Special subtypes

- Tubular carcinoma
- Mucinous carcinoma
- Medullary carcinoma
- Inflammatory carcinoma




Mucinous or Colloid
Carcinoma

LCIS

- Malignant proliferation of cells in lobules
- No invasion of basement membrane
- Discovered incidentally, does not produce mass or calcification
- Dyscohesive cells lacking E- cadherin
- Often multifocal and bilateral

Treatment

- Tamoxifen and close follow –up
- Low risk of progression to invasive carcinoma

Invasive lobular carcinoma

- Grows in single – file pattern
- No duct formation due to lack of E- cadherin

Prognosis based on TNM staging

- Metastasis is most important factor
- Spread to axillary lymph nodes is most useful factor
- Sentinel node biopsy is used to assess axillary lymph nodes

Predictive factors predict response to treatment

- Most important factors are ER, PR and HER2/ neu gene amplification
- ER and PR associated with response to antiestrogen agents (tamoxifen)
- HER2 /neu amplification is associated with response to trastuzumab

Triple - negative

- Negative for ER, PR and HER2/neu
- Poor prognosis
- African American women have an increased propensity to develop TN carcinoma

Hereditary breast cancer

- Represents 10% of cases of breast cancer

Features suggesting hereditary breast cancer

- Multiple – first degree relative with breast cancer
- Tumors at premenopausal age
- Multiple tumors

BRCA1 and BRCA2

- Most important single gene mutations
- BRCA1 breast and ovarian carcinoma
- BRCA2 breast carcinoma in males
- Women with genetic propensity for breast cancer may choose to undergo prophylactic mastectomy

Male breast cancer

- Rare
- Subareolar mass under nipple in older males, may produce nipple discharge
- Usually invasive ductal carcinoma
- Associated with BRCA2 mutations and klinefter syndrome